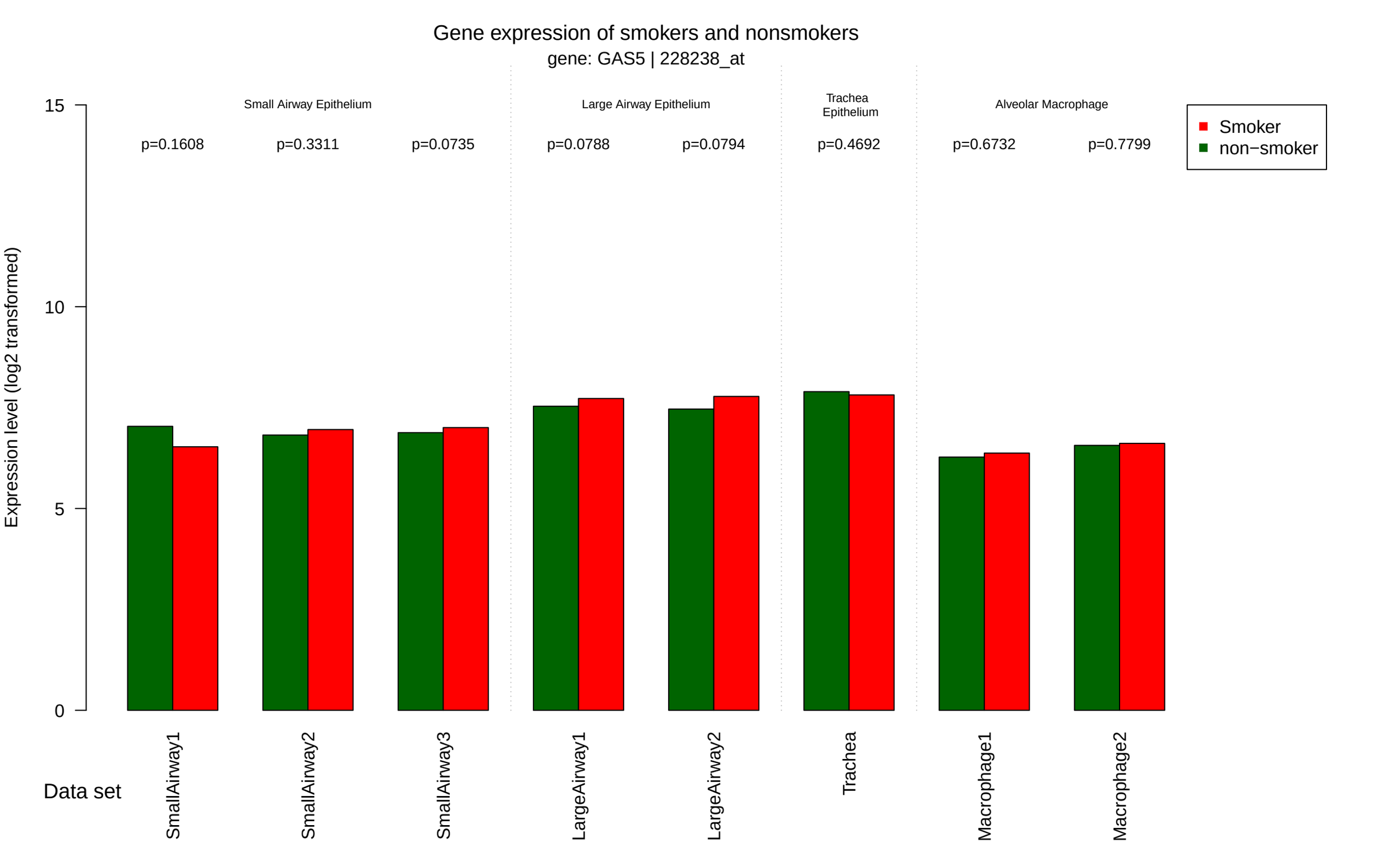
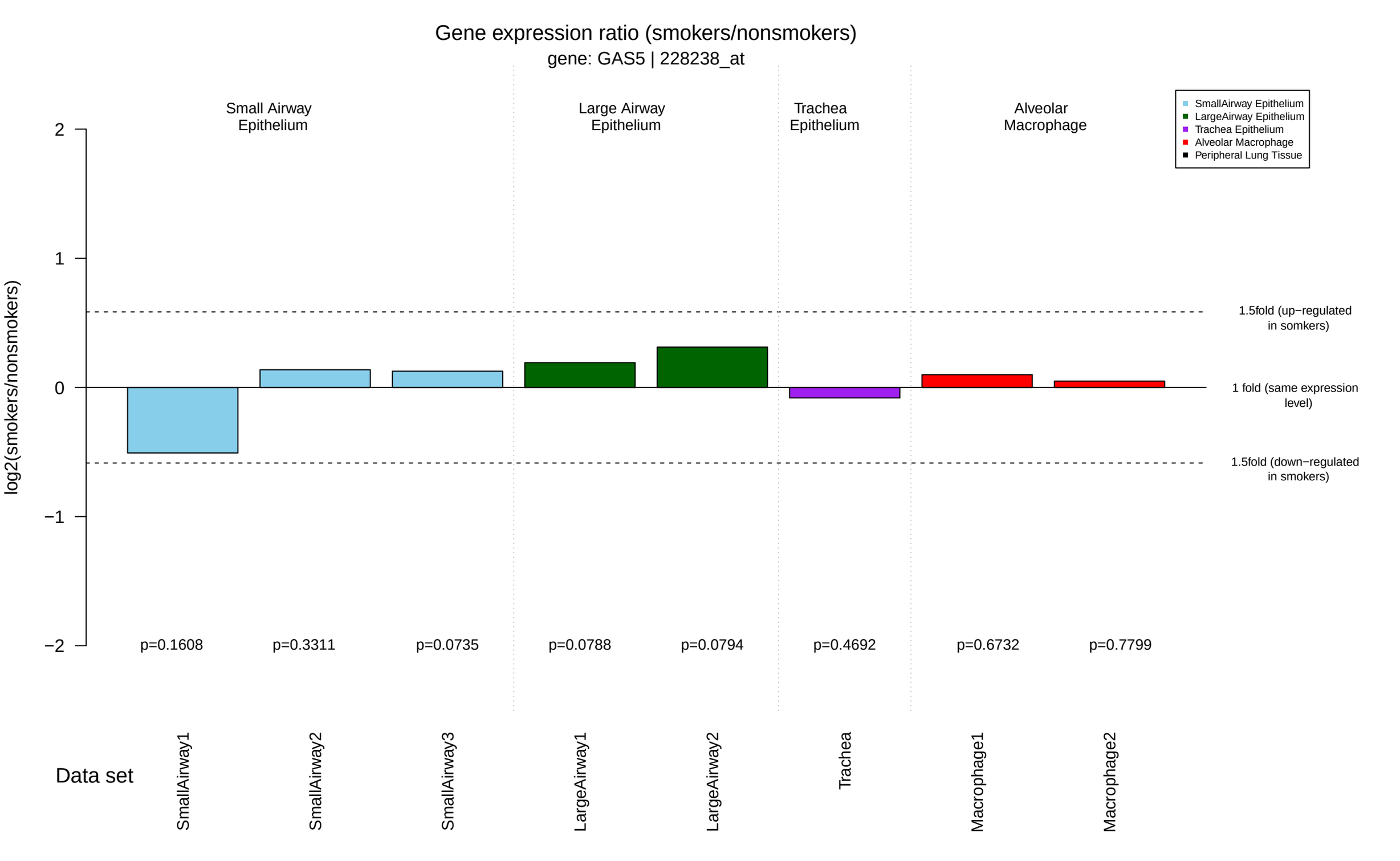
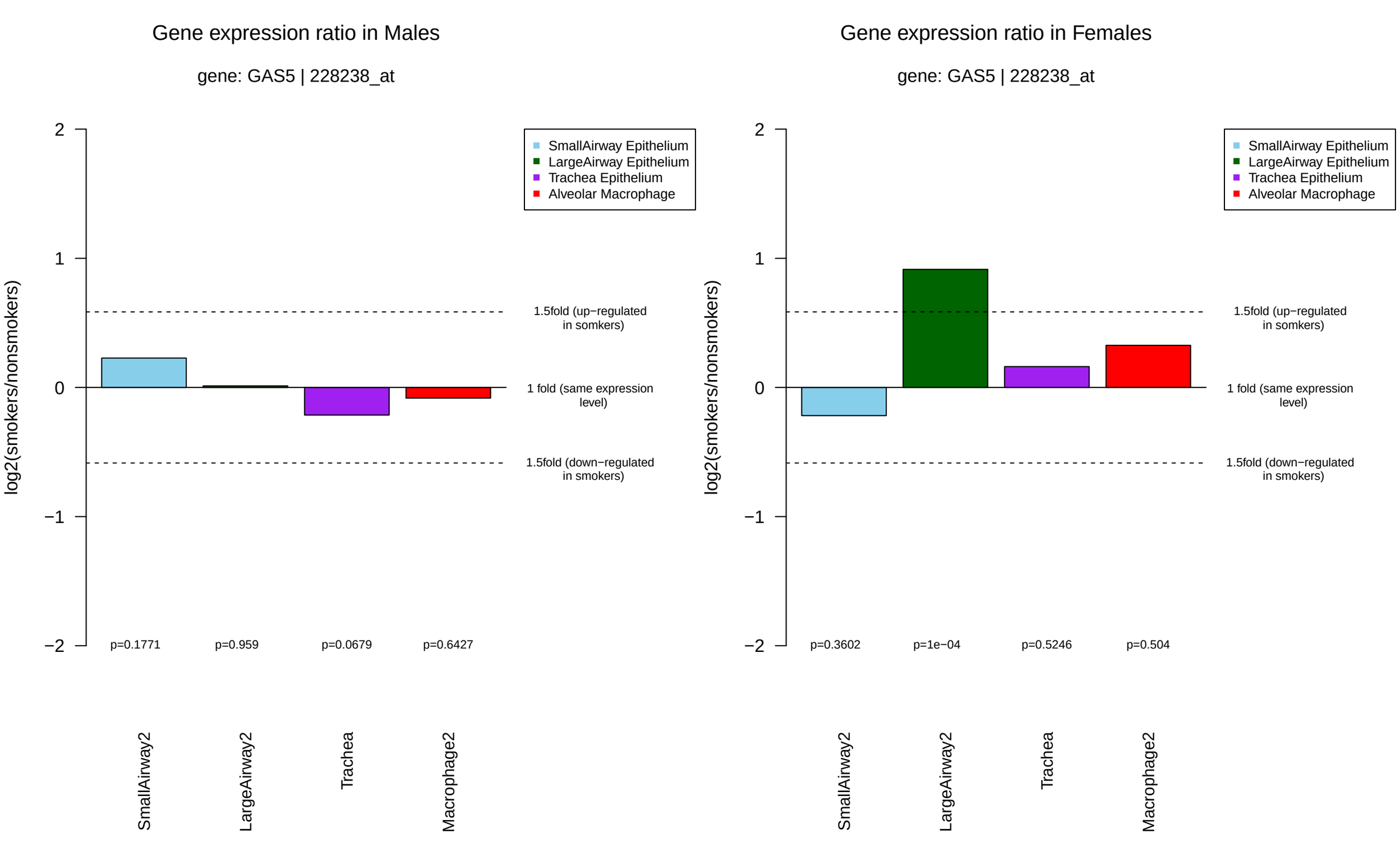
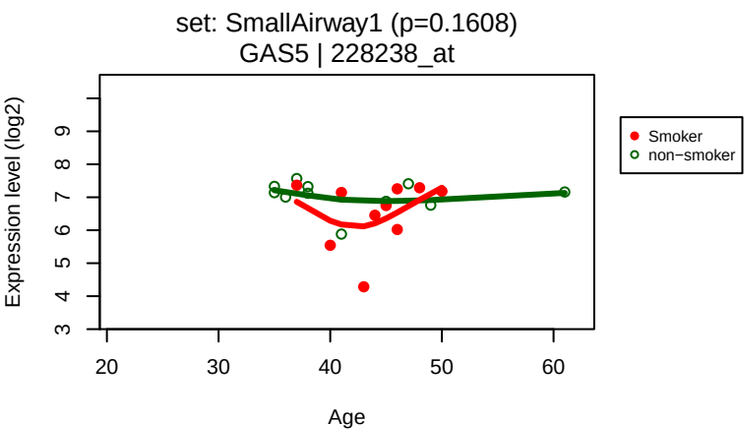
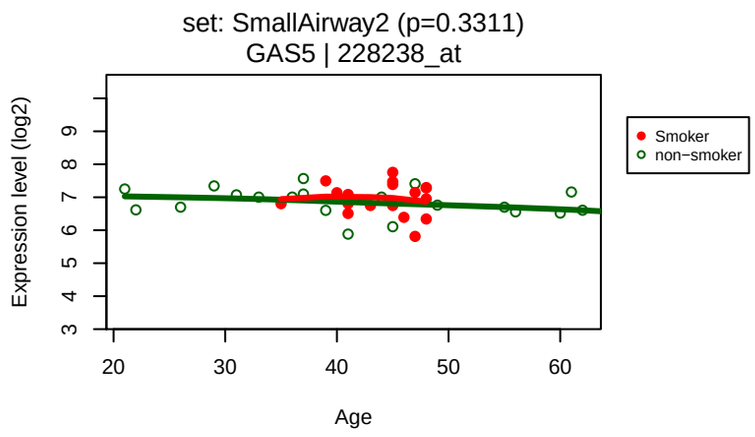
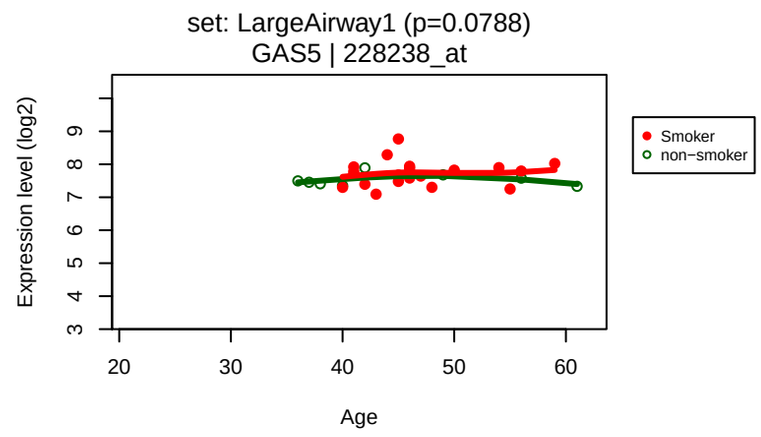

Gene expression of smokers and nonsmokers
gene: GAS5 | 228238_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1608
p=0.3311
p=0.0735
p=0.0788
p=0.0794
p=0.4692
p=0.6732
p=0.7799
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

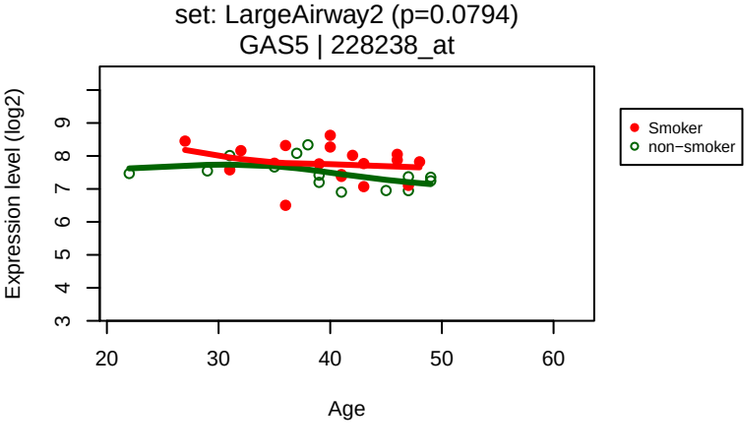

Gene expression ratio (smokers/nonsmokers)
gene: GAS5 | 228238_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1608
p=0.3311
p=0.0735
p=0.0788
p=0.0794
p=0.4692
p=0.6732
p=0.7799
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

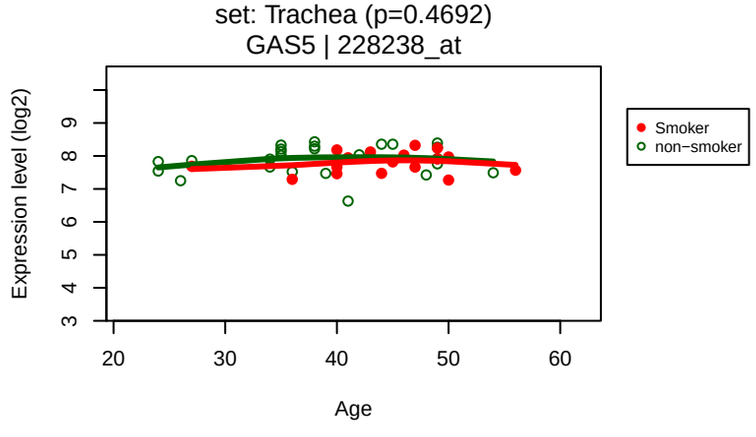

Gene expression ratio in Males
gene: GAS5 | 228238_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1771
p=0.959
p=0.0679
p=0.6427
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GAS5 | 228238_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3602
p=1e−04
p=0.5246
p=0.504
SmallAirway2
LargeAirway2
Trachea
Macrophage2

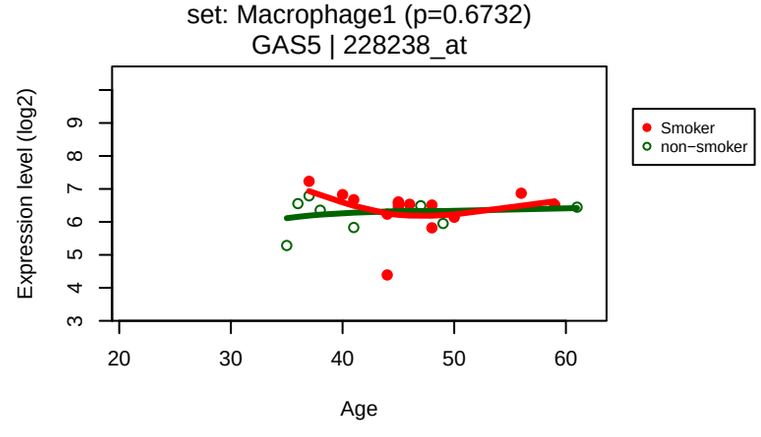

set: SmallAirway1 (p=0.1608)
GAS5 | 228238_at
Expression level (log2)
Age
Smoker
non−smoker

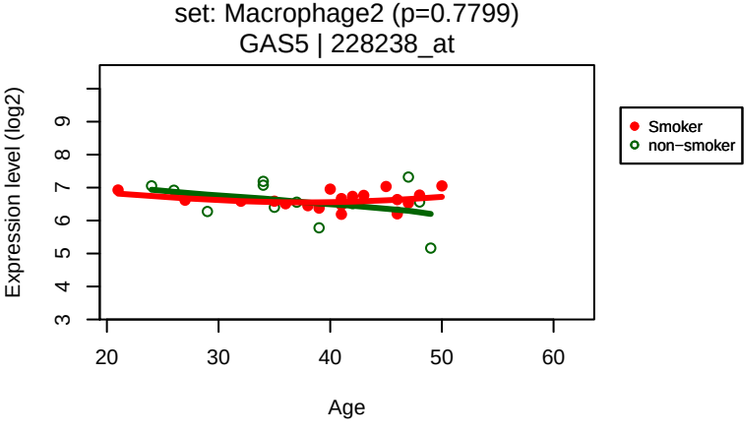

set: SmallAirway2 (p=0.3311)
GAS5 | 228238_at
Expression level (log2)
Age
Smoker
non−smoker

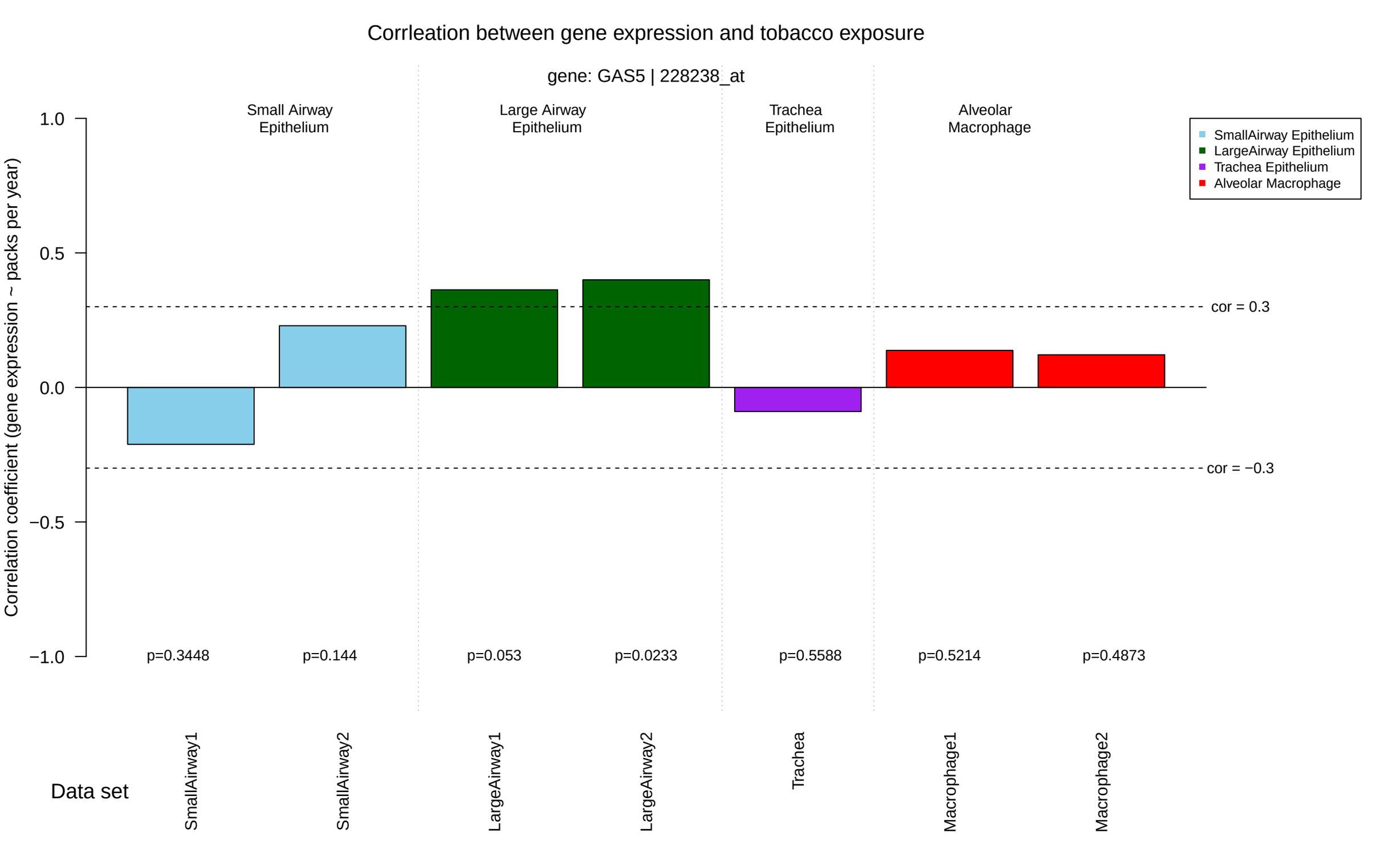

set: LargeAirway1 (p=0.0788)
GAS5 | 228238_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0794)
GAS5 | 228238_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4692)
GAS5 | 228238_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6732)
GAS5 | 228238_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7799)
GAS5 | 228238_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GAS5 | 228238_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.3448
p=0.144
p=0.053
p=0.0233
p=0.5588
p=0.5214
p=0.4873
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2